Draft program. Times, talks and speakers may still change.

2026 Asia-Pacific Solar Research Conference

Open-Source Energy System Modelling Workshop

Monday 30 November 2026
11:00 am for an 11:30 am start • Finishes 4:00 pm, followed by networking
Australian National University, Canberra • In person only • Supported by RACE for 2030

11:00 — 11:30 am	**Arrival, tea & coffee** Informal catch-up and networking before the official start.
11:30 — 11:45 am	**Welcome & introduction** Opening remarks from RACE for 2030, then an overview of the day from the organisers, housekeeping, and an introduction to the all-day activities.
11:45 am — 12:45 pm	**Panel session** Presentations and a panel on ideas, issues and practice in open-source energy modelling, opening up to questions and discussion from the floor. Trustworthy Computational Research in the Age of AI Archie Chapman (UQ)
12:45 — 1:30 pm	Lunch
1:30 — 2:30 pm	**'Short-circuit' talks** Five-minute 'lightning talks' on open-source projects, tools and datasets. NEMX.jl: an open-source Julia package for NEM dispatch modelling Ghulam Mohy Ud Din (CSIRO)
2:30 — 3:30 pm	**Open floor: demos & discussion** 'Short-circuit' presenters host tables for demos, questions and discussion. Explore the projects that interest you.
3:30 — 4:00 pm	**Wrap-up & next steps** Reflections from the day's activities and a discussion of what this community wants to do next.
From 4:00 pm	Networking Informal drinks and conversation.

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Panel session

Trustworthy Computational Research in the Age of AI

A/Prof Archie Chapman, The University of Queensland

Generative AI is rapidly becoming embedded in software- and data-intensive research, creating new opportunities alongside new integrity risks. Drawing on the principles of the Australian Code for the Responsible Conduct of Research, this talk will explore how established practices including verification and validation, benchmarking, open-source software development, data stewardship, and transparent workflows provide a foundation for the responsible use of AI in research. Using examples from energy systems modelling and computational science, I will discuss practical issues such as AI-assisted software development, management of sensitive data, public versus private repositories, software and data licensing, and reproducibility in increasingly automated research environments. The central argument is that AI does not diminish the importance of research integrity principles; rather, it makes rigorous computational research practices more important than ever, while also creating new opportunities for research translation and impact.

'Short-circuit' talks

NEMX.jl: an open-source Julia package for NEM dispatch modelling

Dr Ghulam Mohy Ud Din, CSIRO

NEMX.jl is a Julia package for simulating dispatch in the Australian National Electricity Market. It provides three complementary dispatch models: (1) a zonal model that runs directly on the NEMDE/NEMWEB database and reproduces historical dispatch prices; (2) a nodal model built on the physical Synthetic NEM Network, also driven by NEMWEB data, which reproduces dispatch prices under a range of network formulations from DC through convex relaxations to non-convex AC representations; and (3) lightweight OPF/FCAS models designed for long-term, market-informed planning studies.

The session will give an overview of the package and its capabilities, present benchmark simulations against historical NEM outcomes, and walk through how to set up and run the tool. It will close with a live demonstration of a BESS dispatch investigative study, showing how NEMX.jl can be used to analyze storage behavior and its impact on market outcomes.

Repository: github.com/csiro/ensys-NEMX.jl
Documentation and BESS study: csiro.github.io/ensys-NEMX.jl/dev/bess_study

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